Export of Gigabit Passive Optical Network Encapsulation Mode in IPFIX

draft-netana-opsawg-ipfix-gpon-gem

Enabling **data plane visibility** in passive optical transport of the optical distribution network in broadband access

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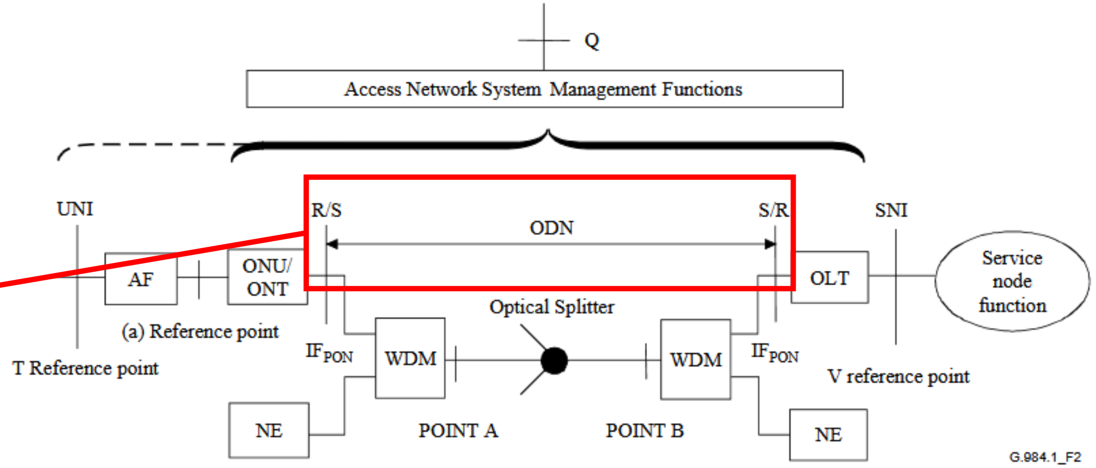
Export of GPON Encapsulation Mode in IPFIX

ITU-T G.984.1 defines the General characteristics

- [ITU-T G.984.1](#) defines the General characteristics of GPON.
- Figure 2 in Section 5.2 shows the Reference Configuration.
- Red marked highlights the Optical Distribution Network (ODN) where GPON Encapsulation Mode (GEM) is used between ONU/ONT and OLT.



ONU	Optical Network Unit
ONT	Optical Network Terminal
ODN	Optical Distribution Network
OLT	Optical Line Termination
WDM	Wavelength Division Multiplex Module (If WDM is not used, this function is not necessary.)
NE	Network Element which uses the different wavelength from the OLT and the ONU
AF	Adaptation Function (Sometimes, it may be included in the ONU.)
SNI	Service Node Interface
UNI	User Network Interface
S	Point on the optical fibre just after the OLT (Downstream)/ONU (Upstream) optical connection point (i.e., optical connector or optical splice)
R	Point on the optical fibre just before the ONU (Downstream)/OLT (Upstream) optical connection point (i.e., optical connector or optical splice)
(a) Reference point	If AF is included in the ONU, this point is not necessary.
POINT A/B	If WDM is not used, these points are not necessary.

NOTE – Whether or not the AF is an operating object of the Q interface depends on the service.

Figure 2 – Reference configuration for GPON

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ITU-T G.984.3 defines the GEM transmission convergence layer

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- Figure 8-11 in Section 8.3.1 shows the GEM header.
- The GEM header contains the payload length indicator (PLI), Port-ID, payload type indicator (PTI) and a 13-bit header error control (HEC) field.
- Red marked highlights the GEM Port-ID and PTI which is of interest to account frames and bytes in IPFIX [RFC 7011], [RFC 7012] and [RFC 7015].
- Payload length indicator (PLI) and header error control (HEC) are for tracing and debugging purposes interesting but not relevant for accounting. Therefore, excluded from IPFIX.

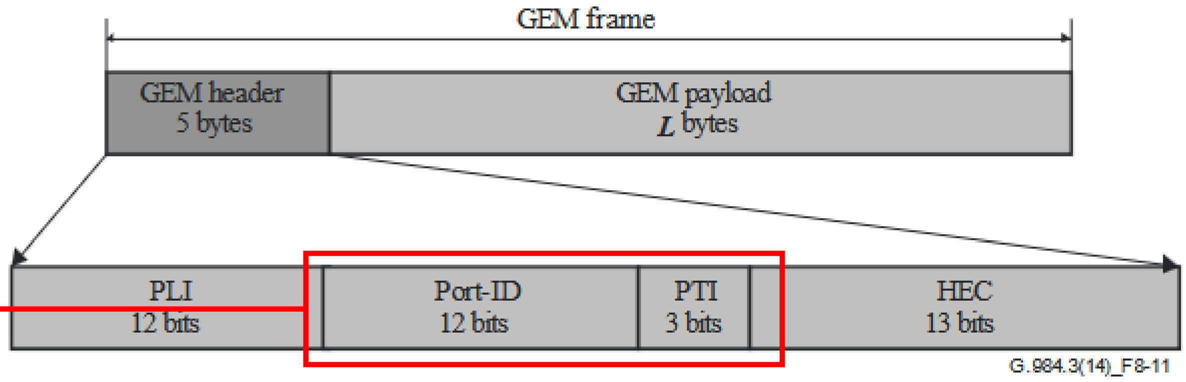

Figure 8-11 – GEM header and frame structure

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ITU-T G.9804.2 defines the XGEM transmission convergence layer

- ITU-T G.9804.2 defines the XGEM transmission convergence layer.
- Figure 8-11 in Section 9.1.2 shows the XGEM header.
- The XGEM header contains the payload length indicator (PLI), Key index, Port-ID, Options, Last Fragment (LF) and Hybrid error correction (HEC).
- Red marked highlights the GEM Port-ID which is of interest to account frames and bytes in IPFIX [RFC 7011], [RFC 7012] and [RFC 7015].
- Payload length indicator (PLI), key index, options, last fragment and header error control (HEC) are for tracing and debugging purposes interesting but not relevant for accounting. Therefore, excluded from IPFIX.

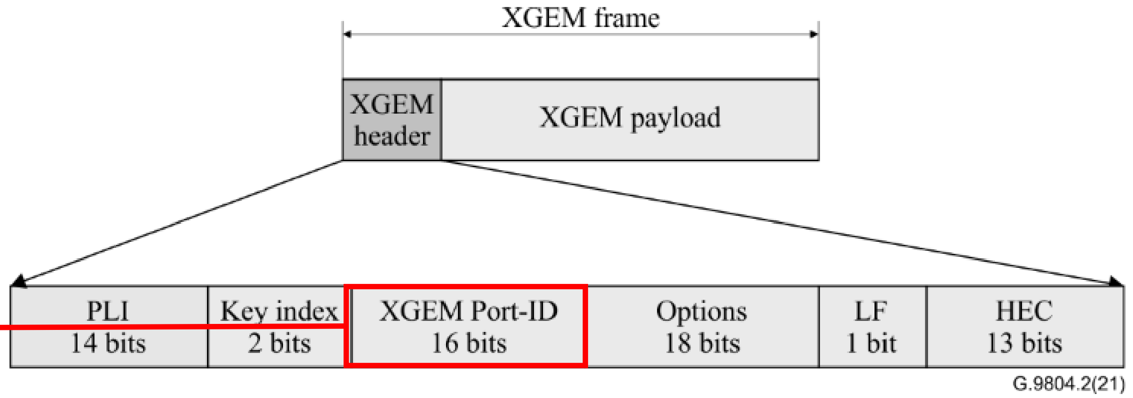

Figure 9-2 – XGEM header format

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draft-netana-opsawg-ipfix-gpon-gem defines IPFIX entities

- **5.1.1. gponGemPti**

Name:	gponGemPti
ElementID:	TBD1
Description:	Values for this Information Element are listed in the Section 8.3.1 of [itu-g984-3]. "G-PON Encapsulation Method PTI" subregistry, see [IANA-IPFIX].
Abstract Data Type:	unsigned8
Data Type Semantics:	flags
Range:	The valid range is 0-7.
Additional Information:	See the assigned types in [IPFIX G-PON content type list. Encapsulation Method PTI Subregistry]. The values encoded in the 3 least significant bits of the IE.
Reference:	[RFC-to-be]

- **5.1.2. gponGemPortId**

Name:	gponGemPortId
ElementID:	TBD2
Description:	The 12-bit GEM Port-ID field defined in Section 8.3.1 of [itu-g984-3] and XGEM Port-ID field in Section 9.1.2 of [itu-g9804-2].
Abstract Data Type:	unsigned16
Data Type Semantics:	identifier
Additional Information:	See Section 8.3.1 of [itu-g984-3] for the content type list.
Reference:	[RFC-to-be]

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1.  Introduction

   The G-PON Encapsulation Method (GEM) data plane header defined in
   Section 8.3.1 of [itu-g984-3] and XG-PON Encapsulation Method (XGEM)
   data plane header defined in Section 9.1.2 of [itu-g9804-2]
   facilitates the framing, error control, encryption keying, payload
   type identification and payload separation in the Optical
   Distribution Network.  It is being used in the Optical Distribution
   Network between the Optical Line Termination (OLT) at the network
   operator and the Optical Network Unit (ONU), Optical Network Terminal
   (ONT) at the end user in the passive optical transport within the
   broadband access domain.

   IPFIX is widely applied in the broadband access domain to gain
   visibility into the forwarding and data plane.  However, that
   visibility is today constraint to the ethernet, IP and application
   transport properties of the data plane.

   This document specifies two IPFIX Information Elements (IEs) to
   facilitate visibility in the GEM and XGEM data plane.
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IPFIX G-PON Encapsulation Method PTI Subregistry

- Based clarifications with Paul Aitken (IE doctor) and Amanda Baber (IANA), the best approach is to mirror the PTI registry from ITU-T G.984.3 as a IPFIX Subregistry.

Value	GEM PTI Content Type Meaning	Additional Information
000	User data fragment, not the end of a frame	[RFC-to-be]
001	User data fragment, end of a frame	[RFC-to-be]
010	Reserved	[RFC-to-be]
011	Reserved	[RFC-to-be]
100	GEM OAM, not the end of a frame	[RFC-to-be]
101	GEM OAM, end of a frame	[RFC-to-be]
110	Reserved	[RFC-to-be]
111	Reserved	[RFC-to-be]

Table 2: "IPFIX G-PON Encapsulation Method PTI" Subregistry

draft-netana-opsawg-ipfix-gpon-gem-01 None

5.1.2.1. G-PON Encapsulation Method PTI Subregistry

This document requests IANA to create a new subregistry called "IPFIX G-PON Encapsulation Method PTI" under the "IPFIX Information Elements" registry [RFC7012] available at [IANA-IPFIX].

The allocation policy of this new subregistry is Expert Review (Section 4.5 of [RFC8126]).

The designated experts for this registry should be familiar with the G-PON Encapsulation Method. The guidelines that are being followed by the designated experts for the IPFIX registry should be followed for this subregistry. In particular, criteria that should be applied by the designated experts include to monitor the G-PON Encapsulation Method related activities at ITU-T and mirror the GEM PTI content type fields into this registry. Hence, keeping both registries in sync.

Initial values in the registry are defined in Table 2 and reflect the 3-bit GEM PTI content type field defined in Section 8.3.1 of [itu-g984-3].

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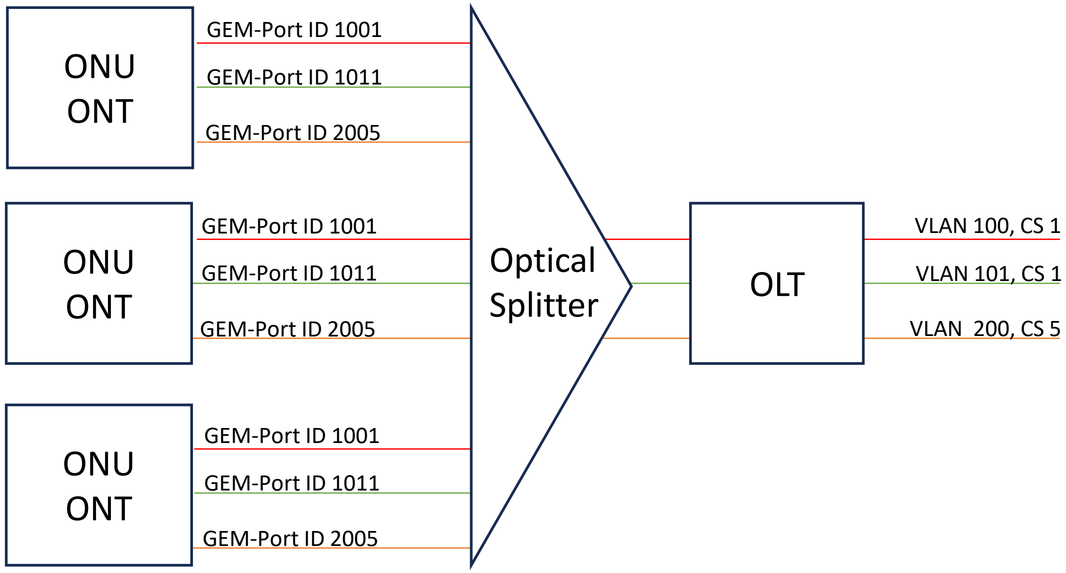
Sample Use Case

GEM Port-ID is usually mapped to IEEE 802.1Q into VLAN identifier (VID) and Priority code point (PCP, IEEE 802.1p class of service) which is used for Network Slicing related use cases. Differentiate between different application and their Quality-of-Service properties.

The IPFIX IEs **gponGemPti (TBD1) or gponGemPortId (TBD2),** sourceMacAddress (56), destinationMacAddress (80), ingressInterface (10), egressInterface (14) and forwardingStatus (89)[RFC5102] [RFC7270] [IANA-IPFIX], and some existing counter information's [IANA-IPFIX] providing answers to the following questions (amongst others):

- **How many user or OAM frames are forwarded or dropped to which ONU on which egress interface and GEM Port-ID?**
- **If dropped, for which reasons?**

The received ONU frames on an OLT are mapped and forwarded depending on GEM Port-ID to a dot1qVlanId (243) and dot1qPriority (244) upstream to the provider network.



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Document Status and Next Steps

- First document revision published on April 7th 2025 and presented at CCAMP.
- Second revision includes review from Paul Aitken and IANA.
- Adopted at OPSAWG September 2025.
- Implementation on Huawei MA5800T-X17 verified at IETF hackathon 125.
- **Received review from Chonfeng Xie and Paul Aitken many thanks!**
- **Addressed review from Chonfeng and working on addressing feedback from Paul**
- **Request working group last call.**

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